Introduction

1. Context & Rationale

The Cognitive Cost of Screen Media

Children spend 5.5–8.6 hours a day on entertainment media — exactly in the years when executive function is built.

Entertainment Surge

All of it outside schoolwork — and rising in every age group.

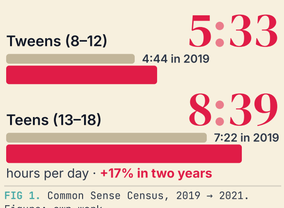

FIG 1. Common Sense Census, 2019 → 2021. Figure: own work.

The Critical Window

The years when reasoning is built are the same years the phone arrives.

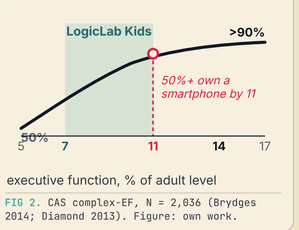

FIG 2. CAS complex-EF, N = 2,036 (Brydges 2014; Diamond 2013). Figure: own work.

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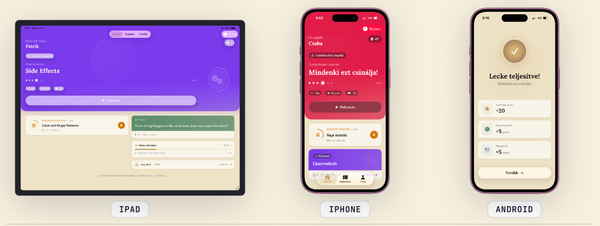

NATIVE SWIFTUI & KOTLIN COMPOSE

OS-ENFORCED REWARD INCENTIVE LOOP

The Premack Behavioral Contingency Model

Premack (1959): a high-probability behavior reinforces a low-probability one. LogicLab Kids makes screen time **earned through reasoning** — enforced by the OS, not by a promise.

REFERENCES

[1] Common Sense Media (2019, 2021). Media Use by Tweens and Teens.

[2] Brydges et al. (2014). Executive function trajectory across ages 5-17 (N=2,036).

[3] Premack, D. (1959). Toward empirical behavior laws: I. Positive reinforcement.

[4] Rideout et al. (2022). The Common Sense Census: Media use by kids.